How Low Can You Go? Lower Limits of Recoverable SMBH Mass in Virgo Cluster UCDs from JWST NIRSpec IFU Data

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Abstract

Ultra-compact dwarf galaxies (UCDs), characterized by their abnormally dense stellar populations and ultra-compact nature, provide a unique tool in the study of galaxy and galaxy group evolution, and possibly exist as a link between various classes of stellar systems (e.g. globular clusters, dwarf ellipticals). Central black holes in such systems not only provide a useful asset in probing the nature of UCDs, but also in the constraint of the black hole mass function in general.

This investigation aims to constrain the lower-limit of detectable M_{BH} mass in compact stellar systems (CSS). This is done to provide preliminary speculations regarding JWST NIRSpec IFU observations of 18 UCDs, NSCs, and cEs, which our data spatially and spectrally mimics. In an effort to best understand how to model our targets, this investigation aims to provide parameterizations of UCDs and cEs in the form of a double power-law spheroidal density model. Using a large sample of *N*-body realizations, we investigate the correlation between observable parameters and those inherent to the density model. This sample is then used to construct accurate systems, which can then be used alongside the FORSTAND code (Vasiliev & Valluri 2020) to investigate the ability to detect the presence of black holes in such systems. We find that systems with $M_{BH}/M_* \lesssim 0.01$ have central black holes whose presence is undetectable with NIRSpec IFU kinematics.

Mock Sample

We constructed a sample of 250 (125 samples each) UCD and cE-like systems in an effort to best determine how one might use observational surface profile parameters to construct three-dimensional *N*-body realizations of such systems. The sample was constructed according to a generalized, three-dimensional $\alpha\beta\gamma$ spheroid model of the form

$$\varrho = \varrho_0 (\acute{r}/a)^{-\gamma} [1+(\acute{r}/a)^{\alpha}]^{(\gamma-\beta)/\alpha} \times \exp[-(\acute{r}/r_{cut})^{\zeta} \quad (1)$$

in which γ, β, and α are the inner powerlaw slope, outer powerlaw slope, and inner-outer transition strength, respectively. The parameters a and r_{cut} are the scale radius and cutoff radius of the system, respectively, while $\acute{r}$, ϱ_0, and ζ are the ellipsoidal radius, density normalization, and cutoff strength of the system. Among the sample, all parameters remained constant with the exception of a and γ. These parameters govern the size and inner density profile of the system, respectively. Each realization was fitted with a two-dimensional Sérsic profile to obtain observable quantities n and r_e.

Two free parameters (γ and a)

↓

Make many mocks over range of suitable γ and a, fix everything else

↓

Project each *N*-body snapshot, fit 2D Sérsic to find n an r_e

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Use observable quantities (n an r_e) to obtain non-observable quantities (γ and a) for each target classification

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Construct mocks based on determined γ and a

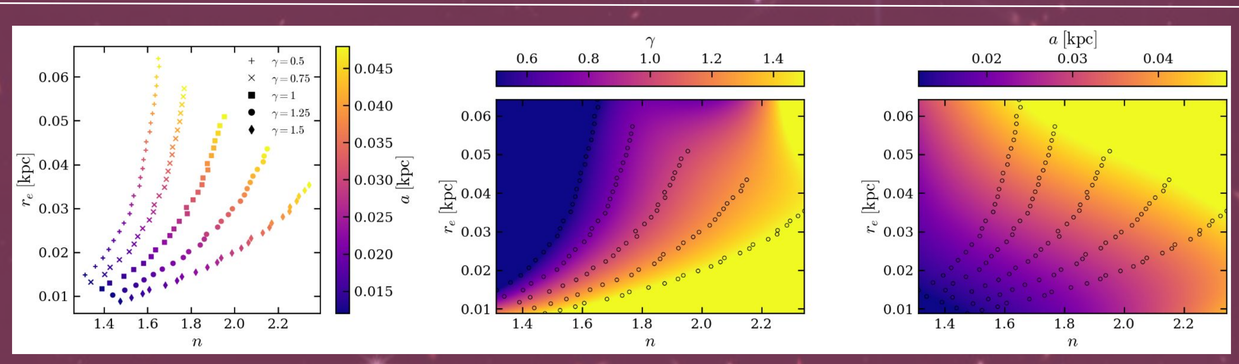

Figure 1: The entire parameter space of the UCD sample. **Left:** Effective radius r_e plotted as a function of Sérsic index n and colored by scale radius a. Each value of γ has its own distinct line. **Center & Right:** The same quantities plotted over a grid of interpolated values via third-order spline interpolation according to either γ (center), or a (right).

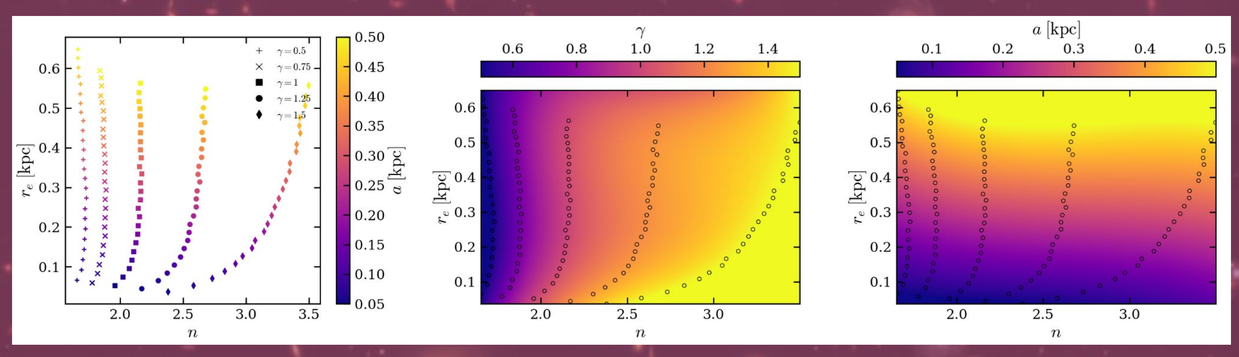

Figure 2: The entire parameter space of the cE sample. **Left:** Effective radius r_e plotted as a function of Sérsic index n and colored by scale radius a. Each value of γ has its own distinct line. Note that, unlike the UCD sample, the cEs have a much stronger correlation between γ and n. **Center & Right:** The same quantities plotted over a grid of interpolated values via third-order spline interpolation according to either γ (center), or a (right).

Example Models

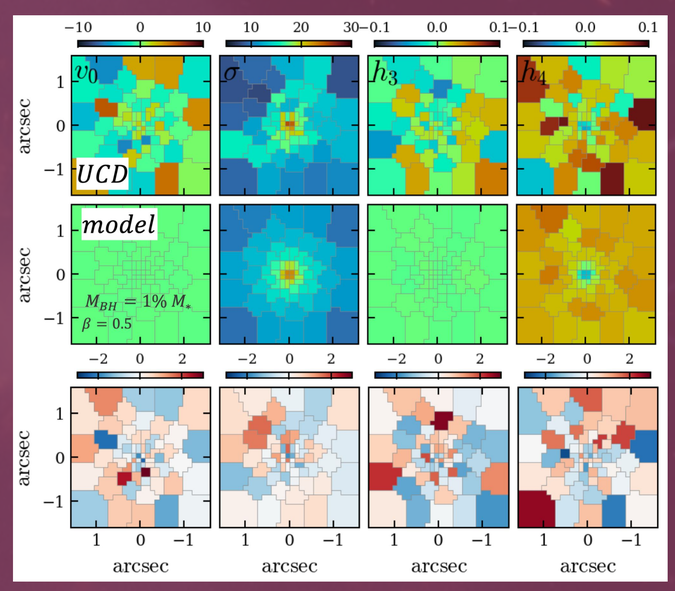

Figure 3: From left to right: line-of-sight velocity, velocity dispersion, and third and fourth order Gauss-Hermite moments. **Top**: Kinematics of a mock UCD-like system generated according to parameters found to be most representative of the classification. The mock has $M_* = 10^{7.4}\ M_\odot$ and central anisotropy of $\beta_0 = 0.5$. **Middle:** The best fit kinematics determined by FORSTAND. **Bottom:** Error in each bin between the actual kinematics and the model.

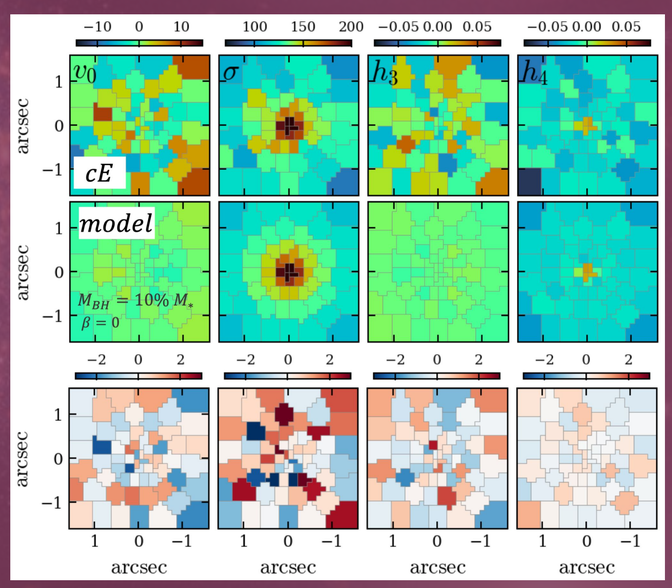

Figure 4: The same quantities as found in Fig. 3, but for a cE-like mock. Note that this example is isotropic (i.e., $\beta_0 = 0$), has a stellar mass component of $M_* = 10^{9.5}\ M_\odot$, and has a considerably higher black hole mass fraction of $M_{BH}/M_* = 0.10$. As one would expect, the extent of the black holes influence is considerably larger than that of the UCD. Also note that, for isotropic systems, h_4 is considerably less prominent outside of the systems central-most region.

Identifying a Lower Limit of Recoverable Mass

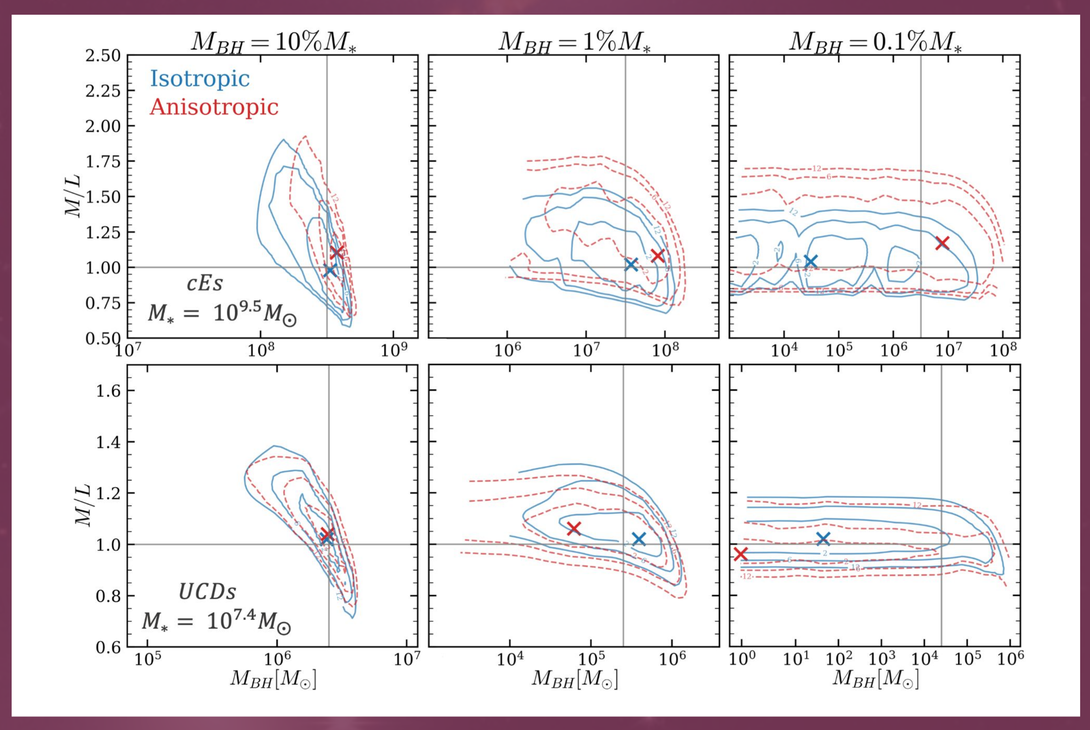

Figure 5: **Top:** The χ^2 plane for cEs with $M_{BH}/M_* = 0.10$, 0.01, and 0.001 respectively. True M/L and M_{BH} values for each mock are denoted with the dashed black cross. For systems with a considerable M_{BH}/M_*, the difference in best-fit values between anisotropic and isotropic realizations is minimal, but this disparity becomes increasingly present for systems with lower M_{BH}/M_*. **Bottom:** The same, but for UCDs. The same behavior between anisotropic and isotropic realizations can be seen, however the realization with the lowest M_{BH}/M_* at the very least provides a reasonable upper limit to M_{BH}.

References

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